A fibration with a point, a decoding map and two equations makes the loop space equivalent to the fibre at the base point

The Univalent Foundations Program

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Editorial note. The counted statement is the author lem environment at source lines 2571-2582 and its proof environment opens on line 2584, the line after the blank separator; the pairing is by adjacency and the retained proof range 2584-2600 is the whole environment.

Editorial note. The pinned book label-number dump carries no entry for this label, so no counter value is imposed and the counted lemma prints with the wrapper numbering; the original number is not claimed.

Editorial note. The statement is a list of four hypotheses whose items carry their own labels (item:ed1 to item:ed4); two of them are cited by the retained proof and both lie inside the retained statement range, so no external reference is needed.

Editorial note. The retained preamble is the book own macro library and layout adapter as stored for this chapter; no package outside it is requested.

Editorial scope. One original result of the pinned HoTT Book source tree, counted once: encode-decode for loop spaces (author label lem:encode-decode-loop).

Allowed standard prerequisites (editorial).

- Transport in a fibration. Along a path between two points of the base, the fibres of a type family are related by transport, and the transport along reflexivity is the identity.. The encoding map is defined as transport of the given point of the fibre at the base point.
- Path induction. To prove a family of statements about all paths out of a point it suffices to prove the case of the reflexivity path.. It reduces the second round-trip identity to a single statement at the base point.
- Quasi-inverses and equivalences. A map is an equivalence as soon as it has a two-sided inverse up to homotopy.. The proof exhibits the decoding map as the inverse of the encoding map on the loop space.

Lemma 0.0.1 (Encode-decode for Loop Spaces)**.** *Given a pointed type* (A, a_0) *and a fibration* $\mathsf{code} : A \to \mathcal{U}$*, if*

(i) $c_0 : \mathsf{code}(a_0)$,
(ii) $\mathsf{decode} : \prod_{(x:A)} \mathsf{code}(x) \to (a_0 = x)$,
(iii) *for all* $c : \mathsf{code}(a_0)$, $\mathsf{transport}^{\mathsf{code}}(\mathsf{decode}(c), c_0) = c$, *and*
(iv) $\mathsf{decode}(c_0) = \mathsf{refl}$,

then $(a_0 = a_0)$ *is equivalent to* $\mathsf{code}(a_0)$.

Proof. Define $\mathsf{encode} : \prod_{(x:A)}(a_0 = x) \to \mathsf{code}(x)$ by

$$\mathsf{encode}_x(\alpha) = \mathsf{transport}^{\mathsf{code}}(\alpha, c_0).$$

We show that encode_{a_0} and decode_{a_0} are quasi-inverses. The composition $\mathsf{encode}_{a_0} \circ \mathsf{decode}_{a_0}$ is immediate by assumption (iii). For the other composition, we show

$$\prod_{(x:A)} \prod_{(p:a_0=x)} \mathsf{decode}_x(\mathsf{encode}_x p) = p.$$

By path induction, it suffices to show $\mathsf{decode}_{a_0}(\mathsf{encode}_{a_o}\mathsf{refl}) = \mathsf{refl}$. After reducing the $\mathsf{transport}$, it suffices to show $\mathsf{decode}_{a_0}(c_0) = \mathsf{refl}$, which is assumption (iv). □